

From Collective Object Transportation to Embodied Coordination: Distributed Cascade Force Control with Minimal Information for Soft-Tactile Multi-Robot Pushing

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Abstract—We propose a novel perspective on collective mobility in multi-robot systems, where a shared object is not only transported but also serves as a physical medium for maintaining coordination among robots capable only of pushing. Each robot is equipped with a unique 360° soft-tactile sensor that estimates the contact force location and magnitude with respect to the object. Through our proposed distributed cascade control strategy with a Lyapunov-based stability guarantee, each robot maintains local pushing contact, becoming implicitly coupled as a loosely connected team moving toward a desired direction. This approach is particularly suitable for unstructured outdoor environments where inter-robot communication, sensing, and global information are limited or unreliable, and object shapes vary. We validate the proposed framework through both simulations and hardware experiments under varying object mass properties and external interferences. The results demonstrate robustness and flexibility in maintaining coordinated motion and achieving object transportation across different object shapes, robot positions, and team sizes. This work highlights its potential for deployment in the wild, across diverse field robotics scenarios.

Index Terms—Distributed control, Lyapunov-based stability, Multi-robot system, Pushing force control, Vision-based tactile sensors

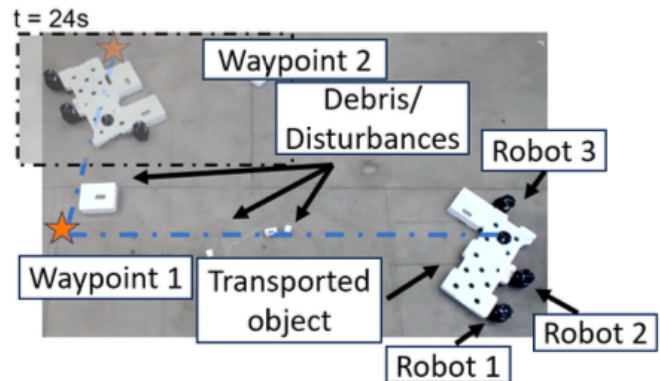


Fig. 1: Three robots push an object or debris toward Waypoint 1, then switch to Waypoint 2 when nearby. Each robot controls its movement and pushing force based on unconventional 360-degree local tactile sensing without knowledge of the object's state, other robots' positions, or their applied forces. This interaction with the shared object indirectly enables the robot team to stay coordinated without requiring relative position information.